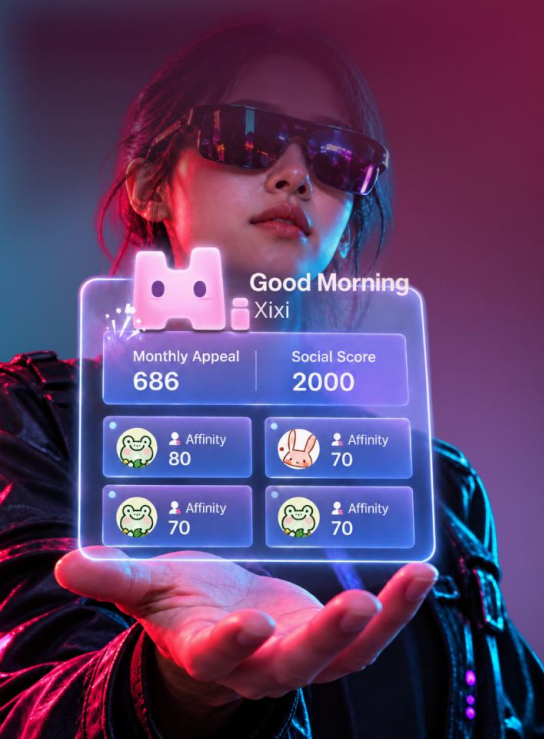




HiMeet: an AR-Assisted Intervention System for People with Social Anxiety

2023 Graduation project



Research

1 DISCOVERY · CAUSAL MODEL

Question

Q1: Why do heavy online socialisers tend to have smaller offline social circles and show more social-anxiety-related behaviours?

Q2: Does social anxiety explain the link between weaker real-world support and greater online social engagement?

User questionnaire statistics



Key Findings

- Weaker real-world social support was associated with higher social anxiety and greater dependence on online socialization.
- Young adults aged 18–25 were especially vulnerable

Design implication: digital support should reconnect people with real-world relationships.

2 PERSONAL SAMPLING · ROLE-PLAY test

Method Self-Report + physiological measures+ video observation

Experimental Procedure Individual session: 45–60 min

1. Read pre-experiment information (2 min) → consent confirmation
2. Social anxiety screening questionnaire (5 min)-LSAS(Liebowitz Social Anxiety Scale)
3. Task & scenario briefing (2 min)
4. Self-perception scale (3 min)
5. Wear EDA wristband for baseline physiological recording (3 min)
6. Situational exposure task (15–20 min)
7. Subjective & somatic self-rating scales (3 min)
8. Semi-structured interview (15 min)



Scenario Tasks

- Enter the room and greet others
- Take a seat
- Introduce yourself appropriately
- Participate in discussion and share opinions
- Initiate one-on-one conversation and exchange contacts

Results

38.0 vs 8.8

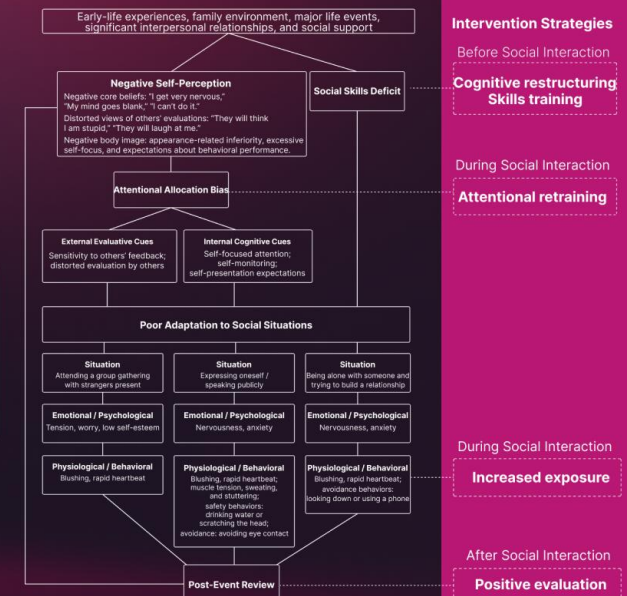
Exposure scenarios is effective: socially anxious participants showed significantly higher subjective distress, $p < .001$

$Y=2.037^*x$

Social anxiety(Y) explained(45.1%) by negative self-cognition(x), especially in negative body-image perceptions, inferiority etc.

Cognitive-behavioral Framework

This test explores cognitions, emotions and behaviours of socially anxious youth in social exposure, and establishes a cognitive-behavioural model, proposing universal intervention strategies targeting frustration triggers.



Design

AR interface design

1 FIRST MEET

The system uses facial recognition to request data from the backend, retrieves personal information, and renders it on the AR device.

Users' social information such as nickname, star sign and occupation is pre-configured and stored in the mobile app.



Try asking about recent TV shows or films he have watched. Be brave and start a new conversation~

BEFORE

Prepare

- Social Classroom
- Cognitive restructuring
- Skills training

MOBILE

DURING

Participate

- Contextual prompts
- Social & emotional cues
- AI for encouraging whisper
- Gesture-based interaction
- Game for encounter

AI+AR

AFTER

Sustain

- Relationship reminders
- Limited messaging online

MOBILE

2 AFTER CHATTING FOR A WHILE

During social interaction, the camera captures facial expressions in real time to recognise emotions. These emotional cues are fed into the AI as auxiliary input for reasoning.



He looks happy. Your conversation is going well !



3 MAKE A CONNECTION

The camera detects human interaction gestures. For instance, it automatically triggers the friend-add action when a handshake is recognised.

Similarly, when the system detects a waving gesture directed toward the user, it displays the prompt: "XX is greeting you; you may try waving back."



Congratulations! You are now friends. Keep in touch~

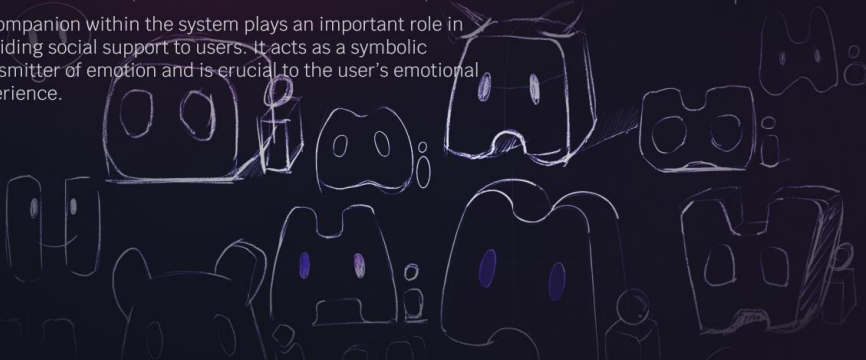


Design

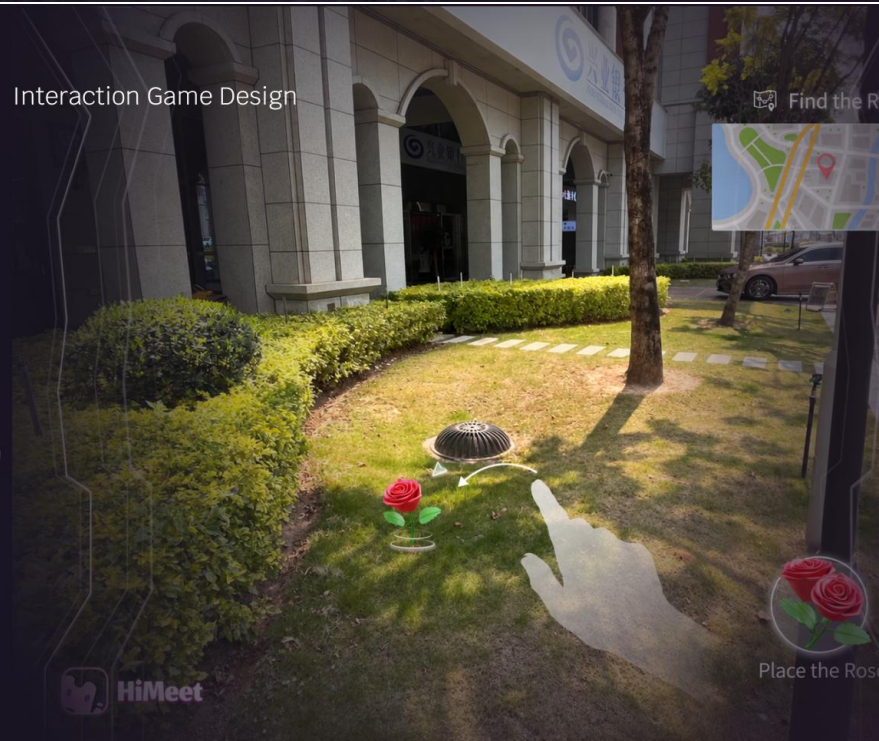
Identity Design



AI companion within the system plays an important role in providing social support to users. It acts as a symbolic transmitter of emotion and is crucial to the user's emotional experience.



Interaction Game Design



"ROSE ENCOUNTER"

an LBS-powered AR interactive game. Users hunt for roses on a digital map. Nearby roses trigger navigation cursors and distance prompts. Tap 3D virtual roses in physical space to collect them into inventory; drag and release the bottom-right icon in the air to place roses. All operations rely on gestures.

